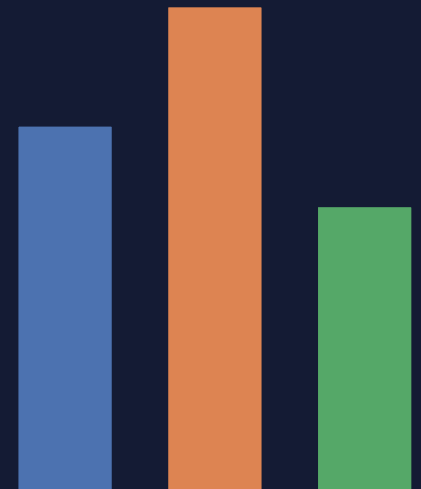
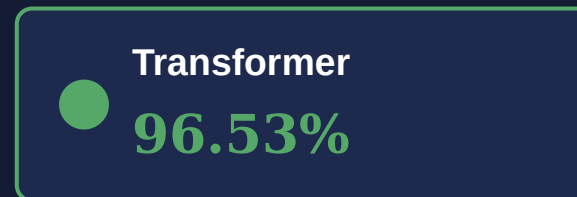


# Recognizing Handwritten Digits

MLP vs CNN vs Transformer on MNIST



## THE EXPERIMENT

# One task, three architectures

MNIST is the *“hello world” of computer vision*: 70,000 handwritten digit images, 0–9. We train on 60,000 and test on 10,000 the model has never seen.

**The question:** does the choice of architecture actually change how well a model reads digits — and how fast it learns?

**70,000**

images total

**5**

epochs each

**Adam**

optimizer + cross-entropy

**64**

images per batch

# How each one sees the image



## MLP

*The baseline*

Flattens the  $28 \times 28$  image into 784 raw numbers, then two dense layers ( $256 \rightarrow 128$ ) with ReLU. Knows nothing about space — just pixel values.



## CNN

*Built for images*

Scans the image with small filters to detect edges and shapes. Two conv layers + pooling shrink  $28 \times 28 \rightarrow 7 \times 7$ , then a small classifier. Understands spatial structure.



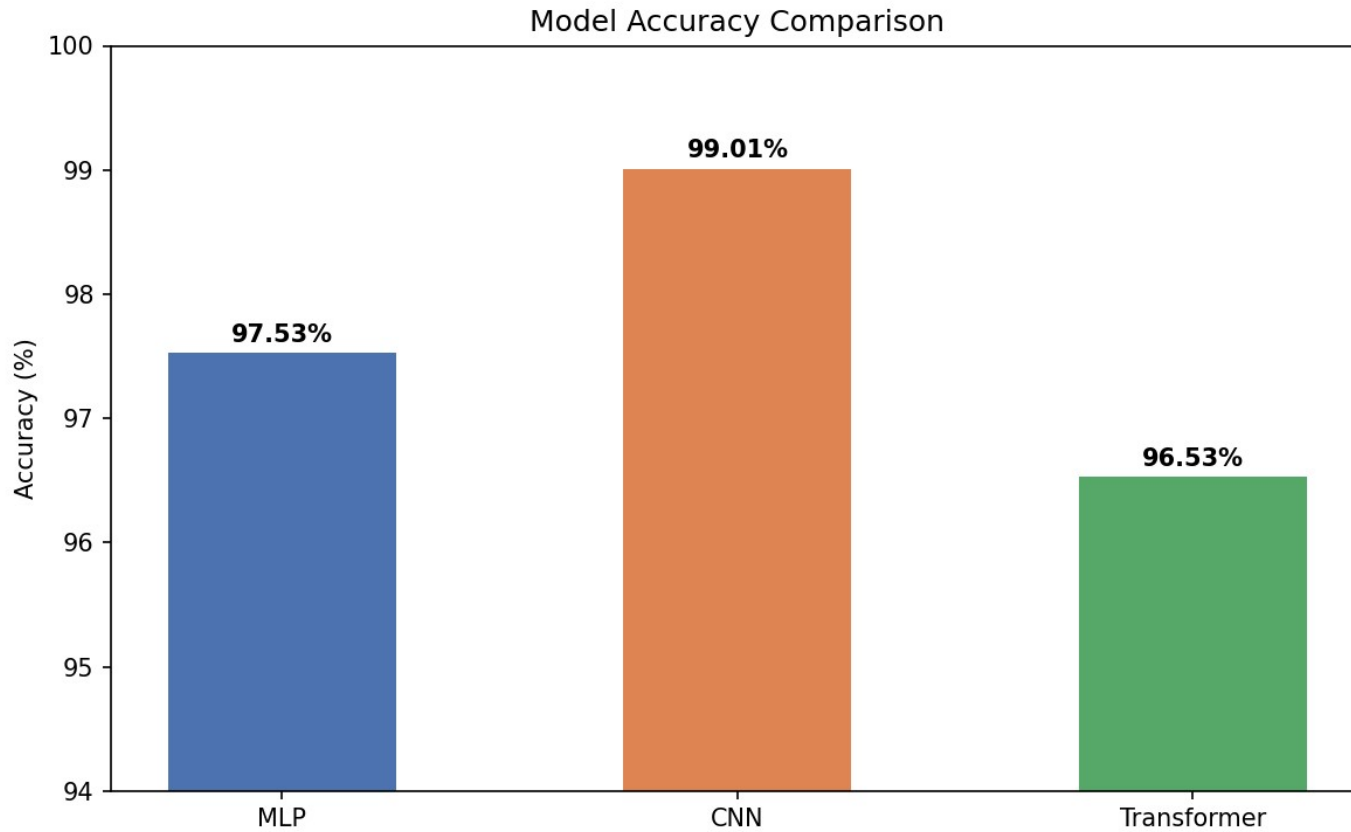
## Transformer

*Borrowed from text*

Splits the image into 16 patches, embeds each, adds a class token + positions, and lets every patch attend to every other via self-attention.

## RESULTS

# CNN wins



Best of the three

# 99.01%

CNN test accuracy

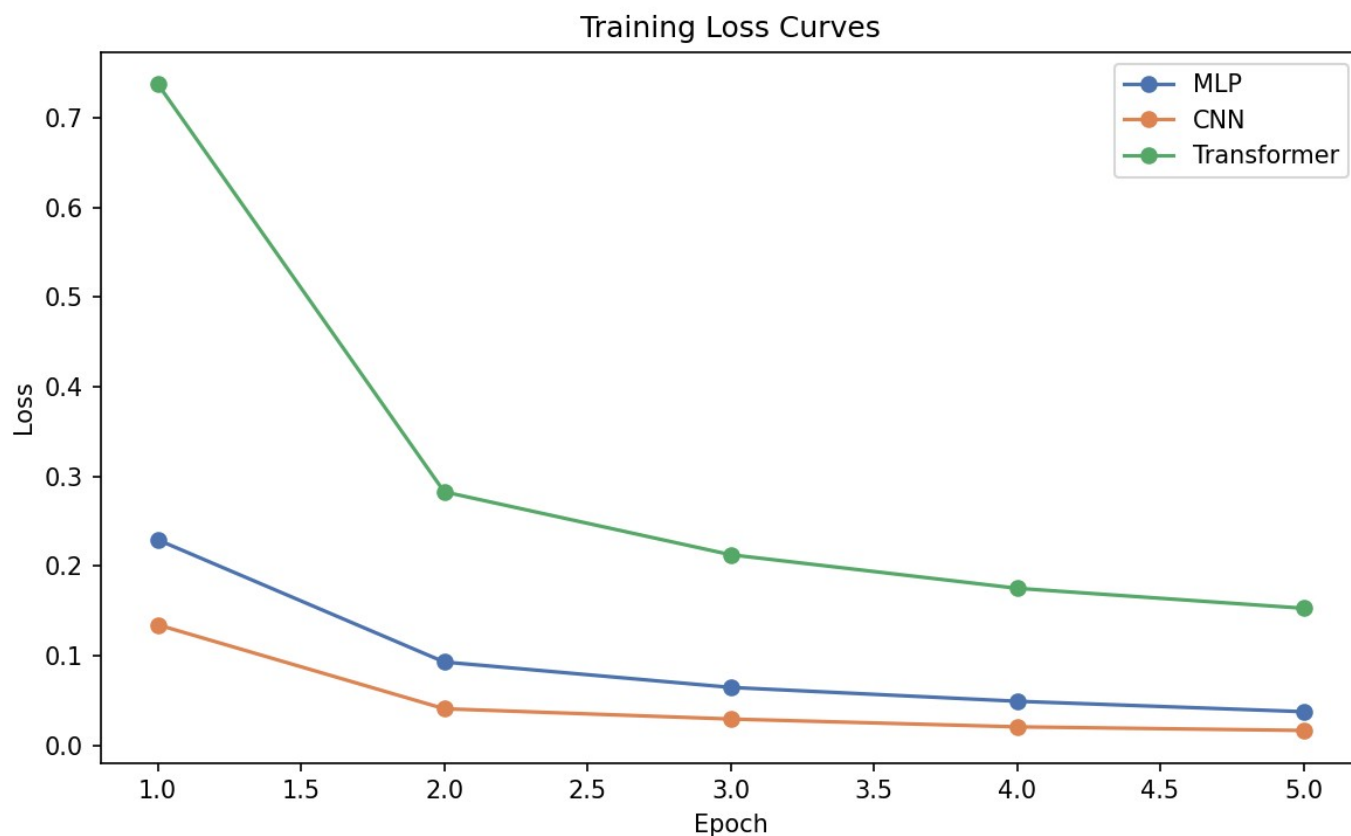
**+1.5 pts** over MLP

**+2.5 pts** over Transformer

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*Scored on 10,000 test images none of the models saw during training.*

# How fast each model learned



## ● CNN

Starts lowest, ends lowest — fits MNIST the fastest.

## ● MLP

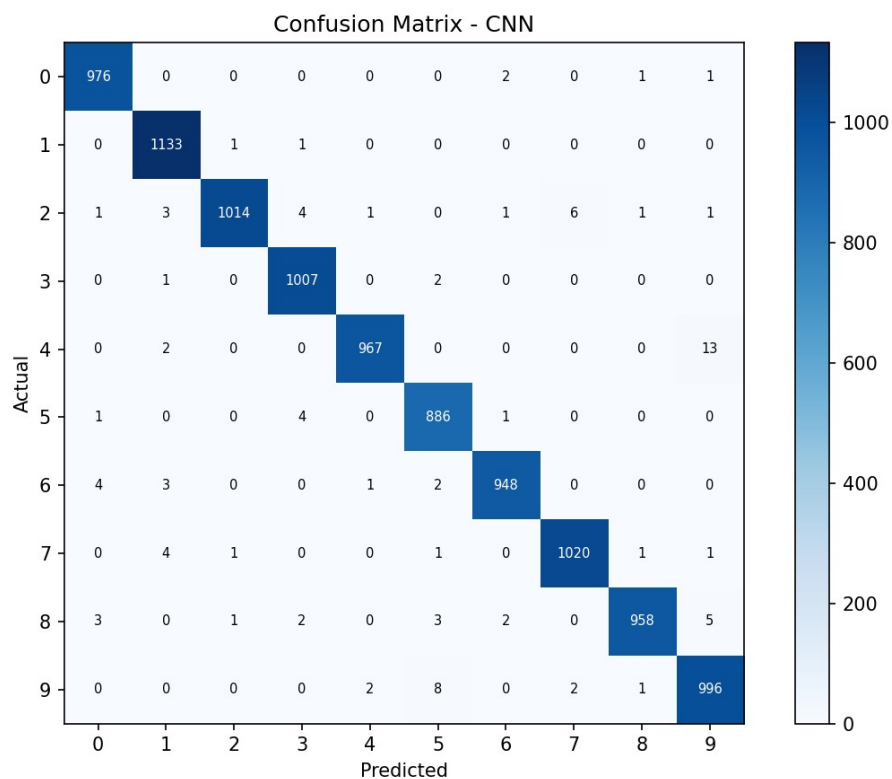
Close behind the CNN; simple but a quick learner.

## ● Transformer

Starts highest (~0.74) and converges slowest — data-hungry and overkill for 28×28 digits.

## ERROR ANALYSIS

# Where the winner slips



Read it as actual (rows) vs predicted (columns). A near-perfect diagonal means almost everything is classified correctly.

**Most common error:** a **4** read as a **9** — 13 times.

The hard pairs — **4/9**, **7/1**, **3/5** — are genuinely ambiguous even to a human eye. All three models trip on the same pairs.

# Three takeaways

1

## Architecture matters.

The CNN wins because convolution is built to understand spatial structure — edges, curves, shapes — not just isolated pixels.

2

## The MLP punches above its weight.

97.5% from raw flattened pixels, with no image-specific tricks at all. A strong, honest baseline.

3

## The Transformer is overkill here.

It works (96.5%), but its strength is large-scale data. On 28×28 digits it learns slowest and gains nothing.